
ForceWAM: A Force-aware Latent World Action Model for Contact-Rich Manipulation

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Abstract

1 Contact-rich manipulation requires precise control and complex physical interaction
2 with the environment, posing significant challenges for robot policy learning.
3 Vision-only policies struggle with such tasks, as contact states and force feedback
4 are difficult to infer from vision alone. Existing methods mitigate this issue by in-
5 corporating force or tactile sensing. However, they often treat these signals as aux-
6 iliary observations rather than using them to model future interaction dynamics,
7 leading to inefficient modality utilization and limited generalization. To address
8 this, we propose ForceLDM, a force-aware latent dynamics model that enables
9 the policy to anticipate future contact and motion dynamics in latent space and
10 use them to guide action generation. Rather than predicting future observations
11 in pixel space, which often requires large amounts of data and risk overfitting ir-
12 relevant details, ForceLDM learns a compact task-relevant representation through
13 knowledge distillation. Specifically, a teacher network with privileged access to fu-
14 ture force/torque and optical flow signals extracts future dynamic features, which
15 are then distilled into a student network trained only on current observations. This
16 enables the student to reason about upcoming contacts and physical dynamics at
17 deployment without relying on future sensory inputs. To improve robustness and
18 generalization, we further introduce a curriculum-based progressive noise injec-
19 tion strategy to mitigate over-reliance on future features. Experiments on five real-
20 world contact-rich manipulation tasks demonstrate that ForceLDM significantly
21 outperforms state-of-the-art baselines, generalizes effectively to novel objects, and
22 remains robust to environmental perturbations.

1 Introduction

24 Humans are naturally capable of performing dexterous manipulation such as USB insertion and
25 component assembly. However, these seemingly simple daily tasks remain highly challenging for
26 robots. Contact-rich manipulation, particularly high-precision insertion, requires not only accurate
27 motion planning and execution, but also the ability to perceive and respond to contact forces in real
28 time, while reasoning over complex physical interaction dynamics. Although vision-based policies
29 have achieved remarkable progress in general-purpose robotic manipulation [19, 4, 3], they still
30 struggle with contact-rich tasks, as contact states and force magnitudes are often difficult to infer
31 reliably from visual alone. This limitation has motivated recent studies to incorporate physical
32 feedback, such as force and tactile sensing, to improve manipulation performance

33 Recent studies on force-aware contact-rich manipulation have primarily focused on designing
34 more effective multimodal architectures to fully exploit force/tactile feedback. For example,
35 ForceVLA [51] employs a Mixture-of-Experts (MoE) architecture with modality-specific experts
36 and uses dynamic routing to adaptively fuse visual and force features. TA-VLA [56] studies joint

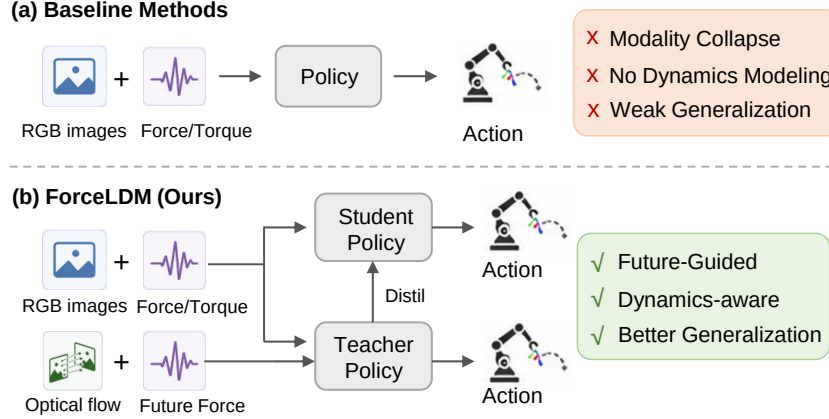


Figure 1: Comparison between baseline methods and ForceLDM. Baselines directly map current observations to actions, which can cause modality collapse and limited generalization, while ForceLDM distills future force and motion representations to make the student learn physical interaction dynamics that guide action generation.

torque fusion in Vision-Language-Action(VLA) models and shows that injecting torque signals into the action generation module outperforms feeding them into the vision-language backbone. Other works [49, 58] investigate fast-slow dual-system architectures, where low-frequency vision provides global scene understanding, while high-frequency force/tactile feedback enables rapid contact response and closed-loop control.

Despite these advances, two fundamental challenges remain unresolved: **modality collapse** and the **lack of physical dynamics modeling**. On the one hand, visual observations typically contain rich global semantics, whereas force/tactile signals are inherently low-dimensional, temporally sparse. Consequently, models tend to over-rely on visual cues for decision-making while largely ignoring or under-utilizing force/tactile feedback a phenomenon we term modality collapse. On the other hand, most existing methods adopt an end-to-end learning framework that directly maps current observations to expert actions without explicitly modeling future physical interaction dynamics. As a result, the learned policy tends to merely imitate expert trajectories rather than comprehend the underlying causality and intent of physical interactions, leading to poor generalization to unseen objects and environmental perturbations.

To address these challenges, we propose **Force-LDM**, a force-aware latent dynamics model for contact-rich manipulation, as illustrated in Figure 1. Force-LDM leverages future force/torque and optical-flow representations as supervisory signals to learn interaction dynamics, which further guide action generation. By introducing future dynamics prediction as an auxiliary objective, Force-LDM requires the model to extract force feedback and motion-relevant cues from current observations, rather than relying solely on visual semantics, thereby mitigating modality collapse and enabling explicit of physical dynamics.

To implement this idea, we employ a knowledge distillation [8, 45] framework. Instead of directly predicting high-dimensional raw future observations, which is computationally expensive and may force the model to capture redundant irrelevant details, we train a privileged teacher model that utilizes both current and future observations to extract compact latent dynamics features, which encode the most useful information about upcoming physical interactions. A student policy is then trained to match these latent representations using only current observations. By distilling such privileged future-aware dynamics into the student, the policy can implicitly capture future interaction patterns without requiring explicit future prediction or future signals at inference time. To further improve generalization, we introduce curriculum-based Progressive Noise Injection, which gradually perturbs the student’s latent queries during training.

We evaluate ForceLDM on five real-world contact-rich manipulation tasks under both in-distribution and out-of-distribution settings. ForceLDM achieves stronger task performance and better generalization to unseen visual and physical variations than existing VLA and force-aware baselines. Fur-

ther ablations and representation analyses show that the student queries encode predictive information about future force and motion dynamics, indicating that ForceLDM implicitly learns physical interaction dynamics rather than merely fitting expert actions. Our contributions are summarized as follows:

- We propose a future physical dynamics representation where force/torque reflects contact states and optical flow encodes motion dynamics, overcoming the limitations of purely visual representations in contact-rich manipulation.
- We introduce ForceLDM, a force-aware latent dynamics model that enables effective physical dynamics modeling from current observations, improving policy robustness and generalization to unseen objects and environmental variations.
- We empirically validate ForceLDM on five real-world contact-rich manipulation tasks, demonstrating its effectiveness in improving task performance, out-of-distribution generalization, and implicit physical dynamics learning.

2 Related Work

Vision-Language-Action Models. Vision-Language-Action (VLA) models [61, 19, 20, 47] have recently emerged as a powerful paradigm for building general-purpose robotic policies [4, 17, 29, 3]. Most VLA models build upon pretrained vision-language models (VLM) [27, 46, 40] and adapt their semantic representations to robotic control. Existing methods mainly follow two action-generation paradigms: discrete autoregressive prediction [33, 19] and continuous action generation [4, 20, 29]. Early methods discretize robot actions into tokens and predict them autoregressively with a VLM, while recent methods attach an action head to generate continuous action chunks through diffusion [6, 13] or flow matching [26, 4]. Building on these foundations, recent studies further improve VLA models by enhancing high-level reasoning and planning [25, 31], strengthening 3D spatial perception [35, 23], incorporating memory mechanisms [38, 44], and developing dual-system architectures [5, 39]. Despite these advances, existing VLA models still struggle with contact-rich manipulation, as vision alone often fails to capture subtle contact dynamics. Therefore, we study force-aware VLA policies that incorporate force/torque signals as physical feedback.

Contact-Rich Manipulation with Physical Feedback. Traditional vision-only methods struggle to capture complex contact dynamics, highlighting the need for integrating physical feedback to improve motion stability and control precision. Current methods rely on three types of physical signals: 6-DoF wrench from force-torque (F/T) sensors [30, 51, 24, 11, 48], which directly measures interaction forces and torques; joint efforts [56, 22, 28], which enable the estimation of external contact dynamics without additional external force sensors; and tactile images [15, 10, 16, 2], which capture detailed, localized geometric profiles of the contact surface. To incorporate these physical signals, existing approaches commonly employ techniques such as feature concatenation [53, 56, 10], cross-attention [12, 60, 21], gating mechanisms [11, 54, 58, 22], or Mixture-of-Experts (MoE) architectures [18, 51, 24] for effective fusion of modalities. However, these methods risk modality collapse by focusing on partial signals and fail to model future contact dynamics, limiting adaptability and generalization. Our approach addresses this by implicitly aligning future force signals and optical flow, effectively modeling contact dynamics and fully utilizing each physical modality.

Predictive Dynamics for Robot Manipulation. Predictive dynamics modeling [14, 7, 55, 43] enables robot learning policies to anticipate the consequences of their actions. Traditional methods typically achieve this using world models [9, 59, 50, 37] that generate future pixel frames to guide planning and control. However, pixel-level prediction is computationally intensive and easily distracted by task-irrelevant appearance changes like lighting and background. While recent latent world models [57, 41, 32] mitigate this by predicting dynamics within abstract representation spaces, they still rely heavily on visual observations. Consequently, these models struggle in contact-rich manipulation, where critical physical events, such as contact onset, slippage, and force buildup, yield minimal visual signatures or are entirely occluded. To overcome these visual limitations, we predict physically grounded signals, including future force-torque measurements and optical flow, as privileged dynamics cues. This encourages the model to learn latent representations that capture contact evolution and motion dynamics, rather than relying on visual similarity, thereby providing direct physical guidance for action generation.

3 Method

In this section, we present **ForceLDM**, a force-aware latent dynamics model for contact-rich manipulation. We begin with the problem formulation in Section 3.1, followed by the core design of ForceLDM in Section 3.2. We then describe future dynamics encoding in Section 3.3, teacher-student distillation in Section 3.4, progressive noise injection in Section 3.5, and the overall training objective in Section 3.6.

3.1 Problem Formulation

We formulate contact-rich manipulation as a policy learning problem. At each time step t , the policy π_θ takes the multimodal observation $o_t = (I_t, s_t, F_t, l)$ as input, and outputs an action chunk $\mathbf{a}_t = (a_t, \dots, a_{t+H-1})$ consisting of H actions to be executed sequentially:

$$\mathbf{a}_t = \pi_\theta(I_t, s_t, F_t, l). \quad (1)$$

where I_t denotes multi-view images, s_t denotes the robot state (end-effector pose or joint positions), F_t denotes 6D wrench measurements, and l denotes the language instruction. To train the policy, we collect a dataset consisting of N expert demonstrations, denoted as $\mathcal{D} = \{\tau_i\}_{i=1}^N$, where each trajectory is defined as $\tau_i = \{(o_t^i, a_t^i)\}_{t=1}^{T_i}$. The policy π_θ is optimized via behavior cloning [1, 36] by minimizing the discrepancy between predicted actions and expert demonstrated actions:

$$\mathcal{L}_{\text{BC}}(\theta) = \mathbb{E}_{(o_t, a_t) \sim \mathcal{D}} [\|a_t - \pi_\theta(o_t)\|_2^2], \quad (2)$$

3.2 Overview of ForceLDM

The high-level idea of ForceLDM is to guide action generation with latent physical dynamics, so that the policy reasons about upcoming contact and motion evolution instead of directly mapping current observations to actions. By learning future force and motion dynamics before predicting actions, the policy is encouraged to attend to force feedback and model physical interaction dynamics rather than relying only on visual semantics. As shown in Figure 2, ForceLDM implements this idea with three modules. First, the future dynamics encoder transforms future force/torque and optical flow into compact force and motion tokens. Second, the teacher-student distillation framework uses these tokens to let a privileged teacher extract future-aware dynamics representations, and aligns the student’s latent queries with the teacher so that the student can infer physical dynamics from current observations. Third, progressive noise injection adds scheduled noise to the student’s latent queries during training, reducing over-reliance on teacher features and improving generalization. At inference time, the teacher and future signals are discarded, and only the student policy is deployed for action generation.

3.3 Future Dynamics Encoding

Choosing the right future modality is critical for contact-rich manipulation. Prior approaches [14, 43, 52] often predict future visual observations, but RGB frames mainly describe appearance and are insufficient to capture physical interaction, such as contact onset, resistance, sliding, or jamming. We therefore use future force/torque measurements and optical flow as privileged dynamics cues. Force directly reflects contact dynamics, while optical flow captures how the robot and objects move through the scene. Together, they provide compact supervision that is more aligned with physical control than reconstructing future RGB frames.

Future Force Encoding. Given a future horizon of K frames, we extract the future 6D wrench sequence $\mathbf{F}_{t:t+K}$ from the force/torque sensor. Instead of using a single force reading, we encode the future force variation over the horizon, which better reflects the temporal evolution of physical contact. The future wrench sequence is projected by an MLP into a force token:

$$\mathbf{z}_t^f = \text{MLP}(\mathbf{F}_{t:t+K}). \quad (3)$$

This token provides the teacher action expert with privileged information about future contact dynamics.

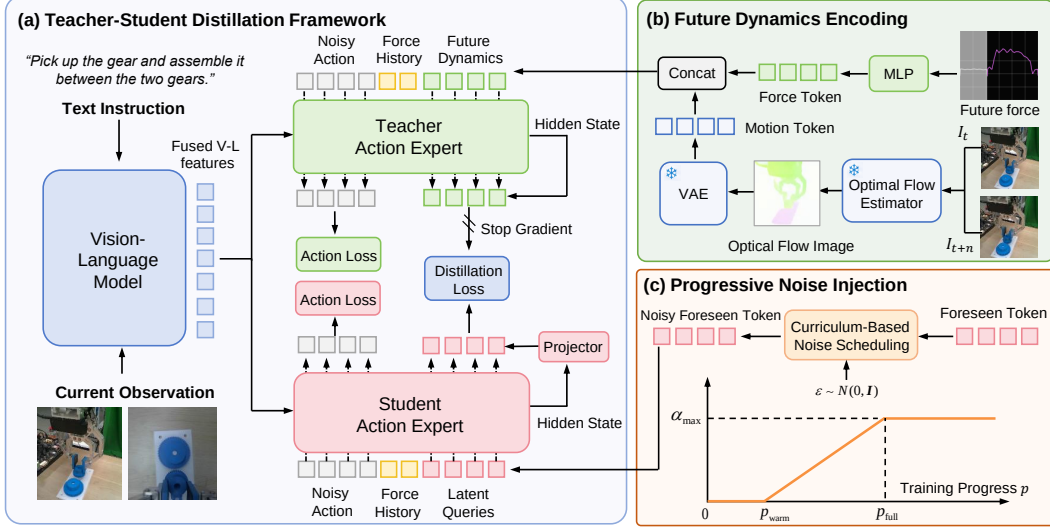


Figure 2: Overview of our proposed ForceLDM. It consists of (a) a teacher-student distillation framework that transfers future-aware dynamics representations to the student policy, (b) a future dynamics encoder that converts future force and optical flow into force and motion tokens, and (c) progressive noise injection that adds scheduled noise to the student’s latent queries to improve generalization.

Future Motion Encoding. Given the current image I_t and a future image I_{t+H} , we use a pre-trained optical-flow estimator, such as RAFT [42], to compute a dense flow field $\mathbf{U}_{t \rightarrow t+H}$. Each pixel has a flow vector $\mathbf{u} = (u_x, u_y)$, which represents its displacement from the current frame to the future frame. To make the flow compatible with image encoders, we convert it into a three-channel image. For each pixel, we compute its angle $\theta = \arctan2(u_y, u_x)$ and magnitude $r = \sqrt{u_x^2 + u_y^2}$. We then form an HSV representation:

$$\mathbf{C}_{t \rightarrow t+H}^{\text{hsv}}(p) = \left(\frac{\theta(p) + \pi}{2\pi}, 1, \text{clip}\left(\frac{r(p)}{\tau}, 0, 1\right) \right), \quad (4)$$

where $\tau = 0.08$ is the normalization scale. We convert this HSV representation into a three-channel RGB flow image $\mathbf{C}_{t \rightarrow t+H}$. We then compress this flow image with a pretrained VAE encoder and use a cross-attention layer with a fixed number of learnable motion queries to extract motion dynamics tokens:

$$\mathbf{z}_t^m = \text{CrossAttn}(Q_m, \text{VAE}(\mathbf{C}_{t \rightarrow t+H})), \quad (5)$$

where Q_m denotes the learnable motion queries, which attend to the VAE-encoded optical-flow features. These tokens provide the teacher with privileged future motion information.

3.4 Teacher-Student Distillation Framework

Distillation allows a privileged teacher to extract action-relevant future dynamics and transfer them to a student policy that only uses current observations. By matching compact latent representations instead of raw future observations, the student learns useful physical dynamics with lower prediction difficulty while avoiding fitting redundant visual details.

Let y denote the teacher’s intermediate future-dynamics query representations induced by privileged force and motion cues, and let h denote the corresponding student latent-query hidden states. In practice, $y \in \mathbb{R}^{L \times N_q \times D}$, where L denotes the number of aligned layers, N_q denotes the number of queries, and D denotes the teacher feature dimension. To align the student with the teacher, we use a lightweight projector f_ϕ to map $h \in \mathbb{R}^{L \times N_q \times D'}$ to y ’s shape. The distillation loss is then defined

190 as:

$$\mathcal{L}_{\text{distill}} = \frac{1}{LN_q} \sum_{l=1}^L \sum_{i=1}^{N_q} [1 - \cos(f_\phi(h^{(l,i)}), \text{sg}(y^{(l,i)}))], \quad (6)$$

191 where $\text{sg}(\cdot)$ denotes stop-gradient. We stop the gradient on the teacher branch to keep the teacher
192 as a stable privileged target. Otherwise, the teacher could also reduce the alignment loss by drifting
193 toward the student, which weakens supervision and may cause representation collapse.

194 3.5 Curriculum Learning with Progressive Noise Injection

195 Inspired by noise injection mechanisms for robust representation learning, we introduce a
196 curriculum-scheduled noise strategy applied to the student’s latent queries. This formulation forces
197 the student policy to anticipate future dynamics from perturbed latent spaces, thereby reducing its
198 reliance on teacher-provided representations.

199 Let $p \in [0, 1]$ denote the normalized training progress, and let $\alpha(p)$ represent the dynamically
200 scheduled noise scale. The perturbed student queries, denoted as $\tilde{\mathbf{Q}}$, are formulated as:

$$\tilde{\mathbf{Q}} = \mathbf{Q} + \alpha(p) \cdot \text{RMS}(\mathbf{Q}) \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (7)$$

201 where $\mathbf{Q}$ denotes the original student latent queries, $\text{RMS}(\cdot)$ computes the root-mean-square statistic
202 to normalize the noise magnitude, and ϵ is a standard Gaussian noise tensor matching the dimensions
203 of $\mathbf{Q}$. We use a three-stage curriculum for $\alpha(p)$, controlled by p_{warm} and p_{full} . Before p_{warm} , no
204 noise is added so the student first learns to align with the teacher; between p_{warm} and p_{full} , the noise
205 scale increases linearly; after p_{full} , it remains fixed at α_{max} .

206 3.6 Training Objective

207 We train both teacher and student action experts with flow matching. Given an expert action chunk
208 a , we sample the same Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$ and time $t \sim \mathcal{U}(0, 1)$ for both branches, and
209 construct the interpolated action $a_t = (1 - t)\epsilon + ta$. The student and teacher predict velocity fields
210 conditioned on c_S and c_T , respectively:

$$\mathcal{L}_S = \mathbb{E}_{a, \epsilon, t} [\|v_S(a_t, t, c_S) - (a - \epsilon)\|_2^2], \quad \mathcal{L}_T = \mathbb{E}_{a, \epsilon, t} [\|v_T(a_t, t, c_T) - (a - \epsilon)\|_2^2]. \quad (8)$$

211 The final objective combines action generation and online distillation:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_S + \mathcal{L}_T + \lambda \mathcal{L}_{\text{distill}}, \quad (9)$$

212 where c_S is conditioned on current observations only, while c_T additionally includes privileged
213 future dynamics. λ balances action learning and physical-dynamics distillation, and is set to 0.5 in
214 all experiments. Since the teacher target is detached in $\mathcal{L}_{\text{distill}}$, the distillation loss only updates the
215 student branch.

216 4 Experiments

217 In this section, we conduct extensive experiments to evaluate the effectiveness of our method on
218 contact-rich manipulation tasks. Our experiments aim to answer the following key questions:

219 **Q1:** Does the proposed force-aware latent dynamics model improve both in-distribution and
220 out-of-distribution performance?

221 **Q2:** How do the proposed future-dynamics representation and progressive noise-injection strat-
222 egy affect performance?

223 **Q3:** Does the learned student representation encode predictive information about future force
224 and optical flow?

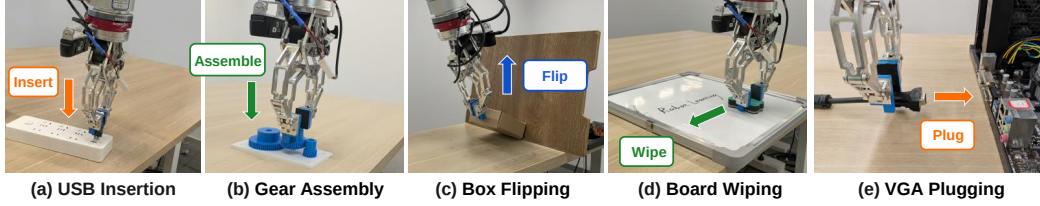


Figure 3: Five real-world contact-rich manipulation tasks used in our experiments.

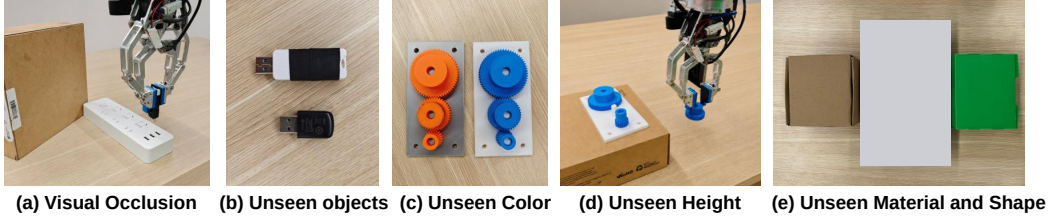


Figure 4: Out-of-distribution generalization settings used in our real-world evaluation.

4.1 Experimental Setup

Hardware Platform and Data Collection We conduct real-world experiments on a 7-DoF xArm7 robotic manipulator equipped with a wrist-mounted 6D force/torque sensor. Visual observations are captured by a wrist-mounted Orbbec Gemini 336L camera and an external fixed RealSense L515 camera. Both data collection and policy execution are performed at 30 Hz. During data collection, an operator uses a 3D SpaceMouse to control the robot, where the relative motion of the SpaceMouse is mapped to the relative motion of the robot end-effector. We record only successful trajectories, including synchronized visual observations, robot states, force-torque readings, and actions, and use them for model training.

Tasks and Evaluation Metrics. We evaluate our method on five real-world contact-rich manipulation tasks: USB Insertion, Gear Assembly, Box Flipping, Board Wiping, and VGA Plugging, as shown in Figure 3. These tasks cover diverse contact-rich interaction patterns, including precise insertion, rigid-body assembly, object reorientation, and sustained surface contact. For each task, we collect 80–100 successful demonstration trajectories and evaluate each method with more than 20 real-world rollouts. We report task success rate as the primary metric. Detailed task descriptions and success criteria are provided in Appendix D.

Compared Baselines. We compare our method against several strong baselines. π_0 [4] **w/o Force** is a flow-matching-based Vision-Language-Action model pretrained on large-scale robot data. π_0 **w/ Force** is a force-augmented variant of π_0 that converts force measurements into tokens and concatenates them with the input sequence. **TA-VLA** [56] is a force-aware VLA model that introduces an auxiliary future-force prediction objective. **ForceVLA** [51] is a force-aware VLA model that adopts a Mixture-of-Experts architecture [18] to model visual and force modalities with specialized experts. **Seer** [43] **w/ Force** is a force-aware variant of Seer that uses force input but directly predicts future visual observations as its auxiliary supervision. For a fair comparison, all methods are fully fine-tuned on our collected dataset.

Implementation Details ForceLDM follows the π_0 architecture, using a pretrained 2.7B PaliGemma VLM and separate 300M teacher and student action experts. The policy predicts action chunks of length 32, uses a 10-frame force history, and encodes future optical flow with an SDXL-VAE [34] followed by cross-attention with 16 learnable motion queries. For distillation, we align action-expert layers $\mathcal{A} = \{8, 12, 16\}$ and train a unified multi-task model for 30,000 iterations. All images are resized to 224×224 with padding before being fed into the vision-language backbone. The initial learning rate is 2.5×10^{-5} , and the minimum learning rate is 1×10^{-6} . Training takes approximately 18 hours on a single NVIDIA A100 80GB GPU.

Table 1: Performance comparison on In-Distribution (ID) contact-rich manipulation tasks. We report the success rate (%) across different tasks. Avg. denotes the average success rate across all ID evaluation settings. Best results are highlighted in **bold**.

Model	USB Insertion	Gear Assembly	Box Flipping	Board Wiping	VGA Plugging	Avg.
π_0 [4] w/o force	44.0	46.7	55.0	60.0	0.0	41.1
π_0 [4] w/ force	64.0	83.3	70.0	70.0	5.0	58.5
TA-VLA [56]	80.0	93.3	80.0	75.0	10.0	67.7
ForceVLA [51]	52.0	66.7	65.0	65.0	5.0	50.7
Seer [43] w/ Force	76.0	70.0	95.0	70.0	50.0	72.2
Ours	92.0	83.3	95.0	80.0	85.0	87.1

Table 2: Generalization performance on Out-of-Distribution (OOD) contact-rich manipulation tasks. We report the success rate (%). Avg. denotes the average success rate across all OOD evaluation settings. Best results are highlighted in **bold**.

Model	USB Insertion		Gear Assembly		Box Flipping		Avg.
	Object	Occlusion	Height	Color	Material	Shape	
π_0 [4] w/o force	20.0	26.7	0.0	40.0	26.7	0.0	18.9
π_0 [4] w/ force	40.0	73.3	0.0	46.7	26.7	33.3	36.7
TA-VLA [56]	20.0	66.7	13.3	66.7	13.3	26.7	34.5
ForceVLA [51]	26.7	20.0	0.0	40.0	0.0	16.7	17.2
Seer [43] w/ Force	46.7	20.0	53.3	60.0	73.3	40.0	48.9
Ours	86.7	86.7	46.7	60.0	80.0	53.3	68.9

4.2 Main Results.

Table 1 summarizes the performance on in-distribution contact-rich manipulation tasks. ForceLDM achieves the best overall performance, demonstrating that modeling future physical dynamics provides effective guidance for contact-rich manipulation. The improvement is most significant on VGA Plugging, where ForceLDM reaches a success rate of 85.0%, compared with 50.0% achieved by the strongest baseline. This result suggests that ForceLDM enables more fine-grained correction during precise insertion. ForceLDM also clearly outperforms Seer, indicating that predicting future visual observations alone is insufficient for high-precision contact-rich manipulation. The main exception is Gear Assembly, where ForceLDM performs worse than TA-VLA. This is mainly because ForceLDM shows limited spatial generalization in this task and sometimes fails to pick the gear at boundary positions.

Training Efficiency. Table 5 compares representative checkpoints on USB Insertion and Gear Assembly. ForceLDM reaches strong performance substantially earlier than TA-VLA: at 5k iterations, it already improves the two-task average by 31.3 percentage points, and by 20k iterations it matches or exceeds the final 30k average of TA-VLA. This faster convergence suggests that distilling future force-and-motion cues provides a more direct learning signal for contact-rich manipulation, reducing the number of optimization steps needed to obtain a reliable policy. Although TA-VLA eventually achieves a higher final success rate on Gear Assembly, ForceLDM maintains better early-checkpoint performance and a higher two-task average across the training process.

4.3 Out-of-Distribution Generalization

OOD Evaluation Settings. We evaluate OOD generalization on three representative contact-rich tasks, as shown in Figure 4. For USB Insertion, we consider *visual occlusion*, where the external camera view is partially blocked by an object, and *unseen objects*, where the USB connector is replaced with a novel instance. For Gear Assembly, we consider *unseen color*, which changes the object appearance, and *unseen height*, which changes the target assembly height. For Box Flipping,

Table 3: Ablation study on future prediction targets. We report success rates (%) on USB Insertion, Box Flipping, and VGA Plugging.

Future Target			Success Rate (%)			
Force	Optical Flow	RGB	USB Insertion	Box Flipping	VGA Plugging	Avg.
–	✓	–	92.0	60.0	60.0	70.7
✓	–	–	84.0	90.0	30.0	68.0
✓	–	✓	80.0	85.0	35.0	66.7
✓	✓	–	92.0	95.0	85.0	90.7

Table 4: Architecture ablation of the proposed teacher–student training design. We report success rates (%) on USB Insertion, Box Flipping, and VGA Plugging. USB Insertion and Box Flipping are evaluated under both in-distribution (ID) and out-of-distribution (OOD) settings. Avg. denotes the average success rate across all reported evaluation settings.

Configuration	USB Insertion		Box Flipping		VGA Plugging	Avg.
	ID	OOD	ID	OOD		
w/o Distill. Loss	44.0	30.0	85.0	30.0	25.0	42.8
w/o Teacher Loss	60.0	36.7	80.0	16.7	10.0	40.7
w/o Stop Gradient	68.0	50.0	60.0	16.7	20.0	42.9
Joint Prediction	84.0	76.7	65.0	20.0	65.0	62.1
w/o Progressive Noise	84.0	73.3	90.0	40.0	85.0	74.1
Full Model	92.0	86.7	95.0	66.7	85.0	85.1

we consider *unseen material*, where the original wooden box is replaced with a 3D-printed box, leading to different friction properties, and *unseen shape*, which evaluates the policy’s ability to flip boxes with different geometries. All OOD settings keep the original task objective and success criterion unchanged, thereby measuring robustness to visual and physical distribution shifts rather than adaptation to new task definitions. Each OOD setting is evaluated over 15 trials.

Results. Table 2 reports the OOD evaluation results. ForceLDM achieves the best average success rate of 68.9%, outperforming the strongest baseline, Seer w/ Force, by 20.0 percentage points. The gains are most pronounced on USB Insertion, where ForceLDM reaches 86.7% under both object variation and occlusion, and on Box Flipping, where it performs best under both material and shape shifts. These results suggest that distilling future force-and-motion cues helps the policy handle changes in object appearance, visibility, and contact response. Gear Assembly shows a more mixed trend: ForceLDM improves over most baselines but remains slightly worse than Seer w/ Force under height changes and TA-VLA under color changes, indicating that fine spatial alignment under visual or geometric shifts remains challenging. Overall, ForceLDM improves OOD generalization by learning future-aware latent dynamics that remain useful beyond the training distribution.

4.4 Ablation Studies

Ablation on Future Prediction Targets. Table 3 studies the role of different future modalities, including force, optical flow, and RGB observations. Future force is particularly useful for tasks that require sustained contact regulation, such as Box Flipping, where force-only supervision already achieves strong performance. Future optical flow provides more direct cues for motion and alignment, which benefits tasks that require repeated local adjustment and probing, such as USB Insertion and VGA Plugging. Combining force and optical flow gives the best overall performance, showing that the two modalities capture complementary aspects of future interaction dynamics. In contrast, adding future RGB provides little benefit and even degrades performance, suggesting that raw visual prediction introduces redundant appearance details that are less action-relevant than optical flow. This supports our design choice of using future force and optical flow as the privileged future representation.

Table 5: Training efficiency comparison at different optimization checkpoints. We report success rates (%) on USB Insertion and Gear Assembly. Avg. denotes the average success rate over the two tasks. Best results at each checkpoint are highlighted in **bold**.

Method	Iter.	USB Insertion	Gear Assembly	Avg.
TA-VLA [56]	5k	–	–	–
	10k	–	–	–
	15k	–	–	–
	20k	–	–	–
	25k	–	–	–
	30k	–	–	–
Ours	5k	–	–	–
	10k	–	–	–
	15k	–	–	–
	20k	–	–	–
	25k	–	–	–
	30k	–	–	–

Ablation on Model Architecture. Table 4 evaluates key components of the proposed teacher–student training design. Removing the alignment loss weakens the transfer of future-aware dynamics from the privileged teacher branch to the deployable student policy, leading to lower overall performance. This result indicates that explicit feature-level distillation is important for enabling the student to acquire predictive physical cues from current observations. We also ablate progressive noise injection, which mainly affects robustness rather than in-distribution fitting: the variant without progressive noise remains competitive on ID settings but performs worse under OOD shifts. This suggests that progressive noise injection prevents the student from overfitting to teacher features and encourages a more stable and transferable future-aware representation. Such regularization is especially important during deployment, where the student must infer future dynamics from current observations alone and may encounter visual or physical conditions not seen during training. Overall, these results show that both explicit teacher–student alignment and regularized distillation are important for learning effective latent dynamics representations.

4.5 Visualization.

Visualization of Latent Dynamics. Beyond aggregate performance, we further analyze whether the learned student queries encode predictive information about future interaction dynamics. We visualize the query representations extracted from frozen policies using a dimensionality-reduction method such as t-SNE or UMAP, and color the resulting embeddings by future force magnitude, contact phase, or task success label. If the proposed alignment mechanism is effective, query embeddings from samples with similar future contact evolution should form more coherent local structure than generic hidden states without alignment. We additionally retrieve nearest neighbors in the learned query space and compare their future force trajectories and future optical-flow patterns. Qualitatively, this analysis can reveal whether the model organizes samples according to anticipated interaction outcomes rather than only current visual similarity. Together with the probe results in Table ??, these visualizations provide complementary evidence that the student queries capture future-aware dynamics relevant to contact-rich manipulation.

5 Conclusion

We presented **ForceLDM**, a force-aware latent dynamics model for contact-rich manipulation. Rather than predicting high-dimensional future observations or simply appending force readings to the policy input, ForceLDM learns a compact future-dynamics representation from privileged future force/torque signals and optical flow. Through teacher–student distillation, this future-aware representation is transferred to a deployable student policy that only uses current observations during inference. We further introduced progressive noise injection to regularize the distillation process and improve the student’s ability to infer future dynamics from causal inputs. Experiments on five real-

world contact-rich manipulation tasks show that ForceLDM improves both in-distribution performance and out-of-distribution generalization over general-purpose and force-aware VLA baselines. Ablation studies further verify the complementary roles of future force and optical flow, as well as the benefit of progressive noise injection. Overall, these results demonstrate that distilling future-aware physical dynamics into latent policy representations is an effective approach to improving action generation in contact-rich robotic manipulation.

References

- [1] Michael Bain and Claude Sammut. A framework for behavioural cloning. In *Machine intelligence 15*, pages 103–129, 1995.
- [2] Jianxin Bi, Kevin Yuchen Ma, Ce Hao, Mike Zheng Shou, and Harold Soh. Vla-touch: Enhancing vision-language-action models with dual-level tactile feedback. *arXiv preprint arXiv:2507.17294*, 2025.
- [3] Johan Bjorck, Fernando Castañeda, Nikita Cherniadev, Xingye Da, Runyu Ding, Linxi Fan, Yu Fang, Dieter Fox, Fengyuan Hu, Spencer Huang, et al. Gr00t n1: An open foundation model for generalist humanoid robots. *arXiv preprint arXiv:2503.14734*, 2025.
- [4] Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, et al. π_0 : A vision-language-action flow model for general robot control. *arXiv preprint arXiv:2410.24164*, 2024.
- [5] Hao Chen, Jiaming Liu, Chenyang Gu, Zhuoyang Liu, Renrui Zhang, Xiaoqi Li, Xiao He, Yandong Guo, Chi-Wing Fu, Shanghang Zhang, et al. Fast-in-slow: A dual-system foundation model unifying fast manipulation within slow reasoning. *arXiv preprint arXiv:2506.01953*, 2025.
- [6] Cheng Chi, Zhenjia Xu, Siyuan Feng, Eric Cousineau, Yilun Du, Benjamin Burchfiel, Russ Tedrake, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research*, 44(10-11):1684–1704, 2025.
- [7] Chelsea Finn and Sergey Levine. Deep visual foresight for planning robot motion. In *2017 IEEE international conference on robotics and automation (ICRA)*, pages 2786–2793. IEEE, 2017.
- [8] Jianping Gou, Baosheng Yu, Stephen J Maybank, and Dacheng Tao. Knowledge distillation: A survey. *International journal of computer vision*, 129(6):1789–1819, 2021.
- [9] Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. *arXiv preprint arXiv:1912.01603*, 2019.
- [10] Peng Hao, Chaofan Zhang, Dingzhe Li, Xiaoge Cao, Xiaoshuai Hao, Shaowei Cui, and Shuo Wang. Tla: Tactile-language-action model for contact-rich manipulation. *arXiv preprint arXiv:2503.08548*, 2025.
- [11] Zihao He, Hongjie Fang, Jingjing Chen, Hao-Shu Fang, and Cewu Lu. Foar: Force-aware reactive policy for contact-rich robotic manipulation. *IEEE Robotics and Automation Letters*, 2025.
- [12] Liang Heng, Haoran Geng, Kaifeng Zhang, Pieter Abbeel, and Jitendra Malik. Vitacformer: Learning cross-modal representation for visuo-tactile dexterous manipulation. *arXiv preprint arXiv:2506.15953*, 2025.
- [13] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.
- [14] Yucheng Hu, Yanjiang Guo, Pengchao Wang, Xiaoyu Chen, Yen-Jen Wang, Jianke Zhang, Koushil Sreenath, Chaochao Lu, and Jianyu Chen. Video prediction policy: A generalist robot policy with predictive visual representations. *arXiv preprint arXiv:2412.14803*, 2024.
- [15] Binghao Huang, Yixuan Wang, Xinyi Yang, Yiyue Luo, and Yunzhu Li. 3d-vitac: Learning fine-grained manipulation with visuo-tactile sensing. *arXiv preprint arXiv:2410.24091*, 2024.

- [16] Jialei Huang, Shuo Wang, Fanqi Lin, Yihang Hu, Chuan Wen, and Yang Gao. Tactile-vla: unlocking vision-language-action model’s physical knowledge for tactile generalization. *arXiv preprint arXiv:2507.09160*, 2025.
- [17] Physical Intelligence, Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, et al. $\pi_{0.5}$: a vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025.
- [18] Robert A Jacobs, Michael I Jordan, Steven J Nowlan, and Geoffrey E Hinton. Adaptive mixtures of local experts. *Neural computation*, 3(1):79–87, 1991.
- [19] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, et al. Openvla: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024.
- [20] Moo Jin Kim, Chelsea Finn, and Percy Liang. Fine-tuning vision-language-action models: Optimizing speed and success. *arXiv preprint arXiv:2502.19645*, 2025.
- [21] Geonhyup Lee, Yeongjin Lee, Kangmin Kim, Seongju Lee, Sangjun Noh, Seunghyeok Back, and Kyoobin Lee. Manipforce: Force-guided policy learning with frequency-aware representation for contact-rich manipulation. *arXiv preprint arXiv:2509.19047*, 2025.
- [22] Jiuzhou Lei, Chang Liu, Yu She, Xiao Liang, and Minghui Zheng. Learning when to see and when to feel: Adaptive vision-torque fusion for contact-aware manipulation. *arXiv preprint arXiv:2604.01414*, 2026.
- [23] Fuhao Li, Wenxuan Song, Han Zhao, Jingbo Wang, Pengxiang Ding, Donglin Wang, Long Zeng, and Haoang Li. Spatial forcing: Implicit spatial representation alignment for vision-language-action model. *arXiv preprint arXiv:2510.12276*, 2025.
- [24] Yang Li, Hongru Jiang, Junjie Xia, Hongquan Zhang, Jinda Du, Yunsong Zhou, Jia Zeng, Ce Hao, Jiejie Ren, Qiaojun Yu, et al. Forcevla2: Unleashing hybrid force-position control with force awareness for contact-rich manipulation. *arXiv preprint arXiv:2603.15169*, 2026.
- [25] Fanqi Lin, Ruiqian Nai, Yingdong Hu, Jiacheng You, Junming Zhao, and Yang Gao. Onetwovla: A unified vision-language-action model with adaptive reasoning. *arXiv preprint arXiv:2505.11917*, 2025.
- [26] Yaron Lipman, Ricky TQ Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. *arXiv preprint arXiv:2210.02747*, 2022.
- [27] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *Advances in neural information processing systems*, 36:34892–34916, 2023.
- [28] Jason Jingzhou Liu, Yulong Li, Kenneth Shaw, Tony Tao, Ruslan Salakhutdinov, and Deepak Pathak. Factr: Force-attending curriculum training for contact-rich policy learning. *arXiv preprint arXiv:2502.17432*, 2025.
- [29] Songming Liu, Lingxuan Wu, Bangguo Li, Hengkai Tan, Huayu Chen, Zhengyi Wang, Ke Xu, Hang Su, and Jun Zhu. Rdt-1b: a diffusion foundation model for bimanual manipulation. *arXiv preprint arXiv:2410.07864*, 2024.
- [30] Wenhai Liu, Junbo Wang, Yiming Wang, Weiming Wang, and Cewu Lu. Forcemimic: Force-centric imitation learning with force-motion capture system for contact-rich manipulation. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 1105–1112. IEEE, 2025.
- [31] Zhuoyang Liu, Jiaming Liu, Hao Chen, Jiale Yu, Ziyu Guo, Chengkai Hou, Chenyang Gu, Xiangju Mi, Renrui Zhang, Kun Wu, et al. Last_{0}: Latent spatio-temporal chain-of-thought for robotic vision-language-action model. *arXiv preprint arXiv:2601.05248*, 2026.
- [32] Hao Luo, Wanpeng Zhang, Yicheng Feng, Sipeng Zheng, Haiweng Xu, Chaoyi Xu, Ziheng Xi, Yuhui Fu, and Zongqing Lu. Being-h0.7: A latent world-action model from egocentric videos. 2026.

- [33] Karl Pertsch, Kyle Stachowicz, Brian Ichter, Danny Driess, Suraj Nair, Quan Vuong, Oier Mees, Chelsea Finn, and Sergey Levine. Fast: Efficient action tokenization for vision-language-action models. *arXiv preprint arXiv:2501.09747*, 2025.
- [34] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. Sdxl: Improving latent diffusion models for high-resolution image synthesis. *arXiv preprint arXiv:2307.01952*, 2023.
- [35] Delin Qu, Haoming Song, Qizhi Chen, Yuanqi Yao, Xinyi Ye, Yan Ding, Zhigang Wang, JiaYuan Gu, Bin Zhao, Dong Wang, et al. Spatialvla: Exploring spatial representations for visual-language-action model. *arXiv preprint arXiv:2501.15830*, 2025.
- [36] Stéphane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the fourteenth international conference on artificial intelligence and statistics*, pages 627–635. JMLR Workshop and Conference Proceedings, 2011.
- [37] Younggyo Seo, Danijar Hafner, Hao Liu, Fangchen Liu, Stephen James, Kimin Lee, and Pieter Abbeel. Masked world models for visual control. In *Conference on Robot Learning*, pages 1332–1344. PMLR, 2023.
- [38] Hao Shi, Bin Xie, Yingfei Liu, Lin Sun, Fengrong Liu, Tiancai Wang, Erjin Zhou, Haoqiang Fan, Xiangyu Zhang, and Gao Huang. Memoryvla: Perceptual-cognitive memory in vision-language-action models for robotic manipulation. *arXiv preprint arXiv:2508.19236*, 2025.
- [39] Haoming Song, Delin Qu, Yuanqi Yao, Qizhi Chen, Qi Lv, Yiwen Tang, Modi Shi, Guanghui Ren, Maoqing Yao, Bin Zhao, et al. Hume: Introducing system-2 thinking in visual-language-action model. *arXiv preprint arXiv:2505.21432*, 2025.
- [40] Andreas Steiner, André Susano Pinto, Michael Tschannen, Daniel Keysers, Xiao Wang, Yonatan Bitton, Alexey Gritsenko, Matthias Minderer, Anthony Sherbondy, Shangbang Long, et al. Paligemma 2: A family of versatile vlms for transfer. *arXiv preprint arXiv:2412.03555*, 2024.
- [41] Yue Su, Sijin Chen, Haixin Shi, Mingyu Liu, Zhengshen Zhang, Ningyuan Huang, Weiheng Zhong, Zhengbang Zhu, Yuxiao Liu, and Xihui Liu. World guidance: World modeling in condition space for action generation. *arXiv preprint arXiv:2602.22010*, 2026.
- [42] Zachary Teed and Jia Deng. Raft: Recurrent all-pairs field transforms for optical flow. In *European conference on computer vision*, pages 402–419. Springer, 2020.
- [43] Yang Tian, Sizhe Yang, Jia Zeng, Ping Wang, Dahua Lin, Hao Dong, and Jiangmiao Pang. Predictive inverse dynamics models are scalable learners for robotic manipulation. *arXiv preprint arXiv:2412.15109*, 2024.
- [44] Marcel Torne, Karl Pertsch, Homer Walke, Kyle Vedder, Suraj Nair, Brian Ichter, Allen Z Ren, Haohuan Wang, Jiaming Tang, Kyle Stachowicz, et al. Mem: Multi-scale embodied memory for vision language action models. *arXiv preprint arXiv:2603.03596*, 2026.
- [45] Lin Wang and Kuk-Jin Yoon. Knowledge distillation and student-teacher learning for visual intelligence: A review and new outlooks. *IEEE transactions on pattern analysis and machine intelligence*, 44(6):3048–3068, 2021.
- [46] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al. Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024.
- [47] Yihao Wang, Pengxiang Ding, Lingxiao Li, Can Cui, Zirui Ge, Xinyang Tong, Wenxuan Song, Han Zhao, Wei Zhao, Pengxu Hou, et al. Vla-adapter: An effective paradigm for tiny-scale vision-language-action model. In *Proceedings of the AAAI conference on artificial intelligence*, volume 40, pages 18638–18646, 2026.

- [48] Yansong Wu, Zongxie Chen, Fan Wu, Lingyun Chen, Liding Zhang, Zhenshan Bing, Abdalla Swikir, Sami Haddadin, and Alois Knoll. Tacdiffusion: Force-domain diffusion policy for precise tactile manipulation. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 11831–11837. IEEE, 2025.
- [49] Han Xue, Jieji Ren, Wendi Chen, Gu Zhang, Yuan Fang, Guoying Gu, Huazhe Xu, and Cewu Lu. Reactive diffusion policy: Slow-fast visual-tactile policy learning for contact-rich manipulation. *arXiv preprint arXiv:2503.02881*, 2025.
- [50] Seonghyeon Ye, Yunhao Ge, Kaiyuan Zheng, Shenyuan Gao, Sihyun Yu, George Kurian, Suneel Indupuru, You Liang Tan, Chuning Zhu, Jiannan Xiang, et al. World action models are zero-shot policies. *arXiv preprint arXiv:2602.15922*, 2026.
- [51] Jiawen Yu, Hairuo Liu, Qiaojun Yu, Jieji Ren, Ce Hao, Haitong Ding, Guangyu Huang, Guofan Huang, Yan Song, Panpan Cai, et al. Forcevla: Enhancing vla models with a force-aware moe for contact-rich manipulation. *arXiv preprint arXiv:2505.22159*, 2025.
- [52] Jia Zeng, Qingwen Bu, Bangjun Wang, Wenke Xia, Li Chen, Hao Dong, Haoming Song, Dong Wang, Di Hu, Ping Luo, et al. Learning manipulation by predicting interaction. *arXiv preprint arXiv:2406.00439*, 2024.
- [53] Chaofan Zhang, Peng Hao, Xiaoge Cao, Xiaoshuai Hao, Shaowei Cui, and Shuo Wang. Vtla: Vision-tactile-language-action model with preference learning for insertion manipulation. *arXiv preprint arXiv:2505.09577*, 2025.
- [54] Kaide Zhang, Heng Zhang, Zhengtong Xu, Zhiyuan Zhang, Md Rakibul Islam Prince, Xiang Li, Xiaojing Han, Yuhao Zhou, Arash Ajoudani, and Yu She. Tacvla: Contact-aware tactile fusion for robust vision-language-action manipulation. *arXiv preprint arXiv:2603.12665*, 2026.
- [55] Wenyao Zhang, Hongsi Liu, Zekun Qi, Yunnan Wang, Xinqiang Yu, Jiazhao Zhang, Runpei Dong, Jiawei He, Fan Lu, He Wang, et al. Dreamvla: a vision-language-action model dreamed with comprehensive world knowledge. *arXiv preprint arXiv:2507.04447*, 2025.
- [56] Zongzheng Zhang, Haobo Xu, Zhuo Yang, Chenghao Yue, Zehao Lin, Huan-ang Gao, Ziwei Wang, and Hao Zhao. Ta-vla: Elucidating the design space of torque-aware vision-language-action models. *arXiv preprint arXiv:2509.07962*, 2025.
- [57] Ruijie Zheng, Jing Wang, Scott Reed, Johan Bjorck, Yu Fang, Fengyuan Hu, Joel Jang, Kaushil Kundalia, Zongyu Lin, Loic Magne, et al. Flare: Robot learning with implicit world modeling. *arXiv preprint arXiv:2505.15659*, 2025.
- [58] Yuhang Zheng, Songen Gu, Weize Li, Yupeng Zheng, Yujie Zang, Shuai Tian, Xiang Li, Ce Hao, Chen Gao, Si Liu, et al. Omnivta: Visuo-tactile world modeling for contact-rich robotic manipulation. *arXiv preprint arXiv:2603.19201*, 2026.
- [59] Chuning Zhu, Raymond Yu, Siyuan Feng, Benjamin Burchfiel, Paarth Shah, and Abhishek Gupta. Unified world models: Coupling video and action diffusion for pretraining on large robotic datasets. *arXiv preprint arXiv:2504.02792*, 2025.
- [60] Xinyue Zhu, Binghao Huang, and Yunzhu Li. Touch in the wild: Learning fine-grained manipulation with a portable visuo-tactile gripper. *arXiv preprint arXiv:2507.15062*, 2025.
- [61] Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, Jialin Wu, Paul Wohlhart, Stefan Welker, Ayzaan Wahid, et al. Rt-2: Vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning*, pages 2165–2183. PMLR, 2023.

Sensitivity to the Alignment Loss Weight. Table 7 studies the influence of the alignment-loss weight λ_{align} on VGA Plugging under the ID setting. A very small weight is insufficient to inject strong future-dynamics supervision into the student queries, while an overly large weight can over-constrain the representation and interfere with action optimization. The best performance is achieved at an intermediate value, indicating that the alignment objective should regularize, rather than dominate, action learning. Overall, the results show that our method is reasonably stable within a moderate range of λ_{align} , while still benefiting from appropriate balancing between action imitation and future-dynamics transfer.

A Limitations and Future Work

Our method has two main limitations. First, ForceLDM relies on an external optical-flow estimator to construct future motion cues during training. Therefore, the learned latent dynamics may be affected by the quality of the estimated flow, especially under motion blur, occlusion. More accurate or task-adaptive motion estimators may further improve the quality of the distilled dynamics representation. Second, our experiments are conducted on a single VLA architecture, namely a π_0 -style policy with a flow-matching action head. Although our framework is architecture-agnostic, its effectiveness on other policy architectures, such as diffusion-based policies, autoregressive action-token models, or other VLA backbones, remains to be validated. Extending ForceLDM to broader robot policy architectures is an important direction for future work.

Broader Impact. Force-aware contact-rich manipulation may benefit robotic assembly, maintenance, and assistive manipulation by improving robustness in tasks where physical feedback is important. At the same time, learned contact policies should be deployed with conservative force limits and safety monitors, since incorrect contact reasoning can damage objects, fixtures, or robot hardware.

B Data Collection

Figure 5 shows the real-world experimental setup and representative task initializations used for data collection and evaluation. Before each rollout, we reset the robot, manipulated object, and task fixture to the corresponding initial configuration, and verify that the target region is visible from the camera views. The demonstrations are collected by teleoperation with the same workspace layout and sensing setup used during policy evaluation. For each task, the initial pose preserves the intended contact-rich challenge while avoiding unreachable or unsafe starting states. During both demonstration collection and policy evaluation, the robot starts from the same task-specific reset region, and the manipulated object and target fixture are placed according to the corresponding initialization shown in the figure.

Dataset Statistics. Our collected dataset contains 420 successful demonstration trajectories in total. The per-task trajectory counts are summarized in Table 6.

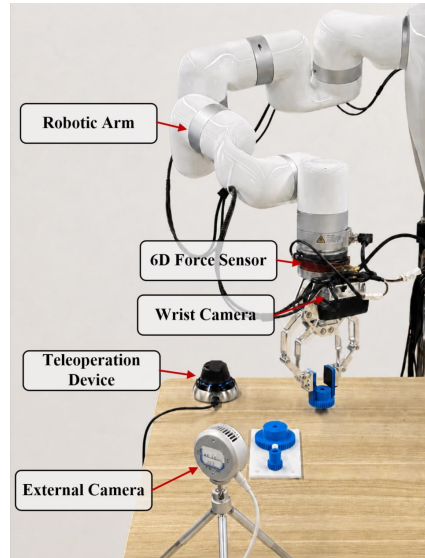


Table 6: Number of successful demonstration trajectories in our collected dataset.
Figure 5: Real-world experimental setup and task initialization examples.

Task	Number of Trajectories
USB Insertion	100
Gear Assembly	80
Box Flipping	80
Board Wiping	80
VGA Plugging	80
Total	420

C Details of Compared Baselines

Seer+Force. This baseline tests whether the Seer architecture can benefit from force sensing through direct multimodal fusion alone. We augment Seer with encoded 6-axis wrist force/torque tokens and concatenate them with

the original multimodal inputs before action prediction. The training objective is kept unchanged, so this comparison isolates the effect of adding force observations without introducing structured future-dynamics supervision.

π_0 w/o force. We adopt π_0 as the primary generic VLA baseline because it is a strong pretrained vision-language-action policy with a flow-matching action head. For our experiments, π_0 takes the same visual observations, language instruction, and robot proprioceptive state as our method, but it does not receive force measurements and does not include any explicit future-dynamics modeling module. The model is fine-tuned end-to-end on our task-specific demonstrations and predicts the same action-chunk format as all other methods.

π_0 w/ force. This baseline tests whether simply exposing a strong VLA model to force signals is sufficient for contact-rich manipulation. We extend π_0 by encoding the recent 6-axis force/torque history into force tokens and concatenating them with the multimodal input sequence. The action head and training objective remain unchanged. This comparison isolates the benefit of force observation alone from the benefit of structured predictive dynamics learning.

TA-VLA. TA-VLA is included as a representative force-aware VLA baseline that uses an auxiliary future-prediction objective during training. In our setting, the model consumes the same causal inputs as our method and is fine-tuned on the same dataset split. Unlike our method, TA-VLA does not decompose policy learning into separate semantic planning, force prediction, and action generation stages. Instead, future-force supervision is used as an auxiliary signal to regularize a largely monolithic policy.

ForceVLA. ForceVLA serves as a stronger multimodal baseline with modality-specialized experts. It uses separate experts to process visual and force inputs and fuses them for action prediction. We keep the observation interface, action representation, and fine-tuning protocol consistent with the rest of the baselines. This comparison helps determine whether gains come simply from larger multimodal capacity or from explicitly modeling future contact dynamics.

D Task Descriptions and Success Criteria

USB Insertion. The robot grasps a USB connector, aligns it with the target port, and inserts it fully into the receptacle. This task requires accurate pose alignment, contact detection at the port entrance, and small corrective motions during insertion. A rollout is counted as successful if the USB plug enters the port with the correct orientation and reaches the fully seated configuration without being manually corrected or forcefully wedged at an incorrect angle. Failed partial insertions, obvious misalignment, or unstable contact that cannot progress to full insertion are counted as failures.

Table 7: Sensitivity analysis of λ_{align} on VGA Plugging.

λ_{align}	VGA
0.1	–
0.2	–
0.5	–
1.0	–

Gear Assembly. The robot picks and places a gear onto a target shaft or matching fixture and completes the mating process under contact constraints. The main challenge is resolving small pose errors during the final alignment stage, where the gear teeth and the shaft geometry create narrow feasible contact transitions. A rollout is successful if the gear is assembled into the designated target position and remains stably seated after release. Cases with visible tilting, incomplete seating, or contact deadlock are counted as failures.

Box Flipping. The robot reorients a box-like object by establishing and maintaining controlled contact with the table or another support surface. The task requires coordinated force modulation to prevent slipping or losing the object while transitioning between support modes. A rollout is successful if the object reaches the specified target orientation and remains stable at the end of execution. Any trial with object drop, uncontrolled slip, or incorrect final orientation is labeled as failure.

626 **Board Wiping.** The robot moves a tool or contact surface across a board while maintaining sta-
627 ble contact and appropriate motion coverage. Unlike insertion tasks, this task emphasizes sustained
628 surface interaction rather than a single terminal contact event. A rollout is successful if the robot
629 preserves continuous task-relevant contact with the board and completes the prescribed wiping mo-
630 tion over the target region without losing contact or deviating substantially from the intended path.
631 Rollouts with premature lift-off, unstable chatter, or incomplete coverage are counted as failures.

632 **VGA Plugging.** The robot aligns a VGA connector with the socket and performs a plug-in mo-
633 tion under tight geometric tolerance and strong contact sensitivity. Compared with USB insertion,
634 this task typically involves a wider connector body and stricter alignment requirements, making fail-
635 ure recovery more difficult once the connector is skewed. A rollout is successful if the connector
636 is inserted into the VGA port to the target depth and remains securely engaged at the end of the
637 rollout. Misaligned insertion attempts, shallow engagement, or jamming outside the valid socket
638 configuration are counted as failures.

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